Implement Snowflake Streams for Sales Data Processing

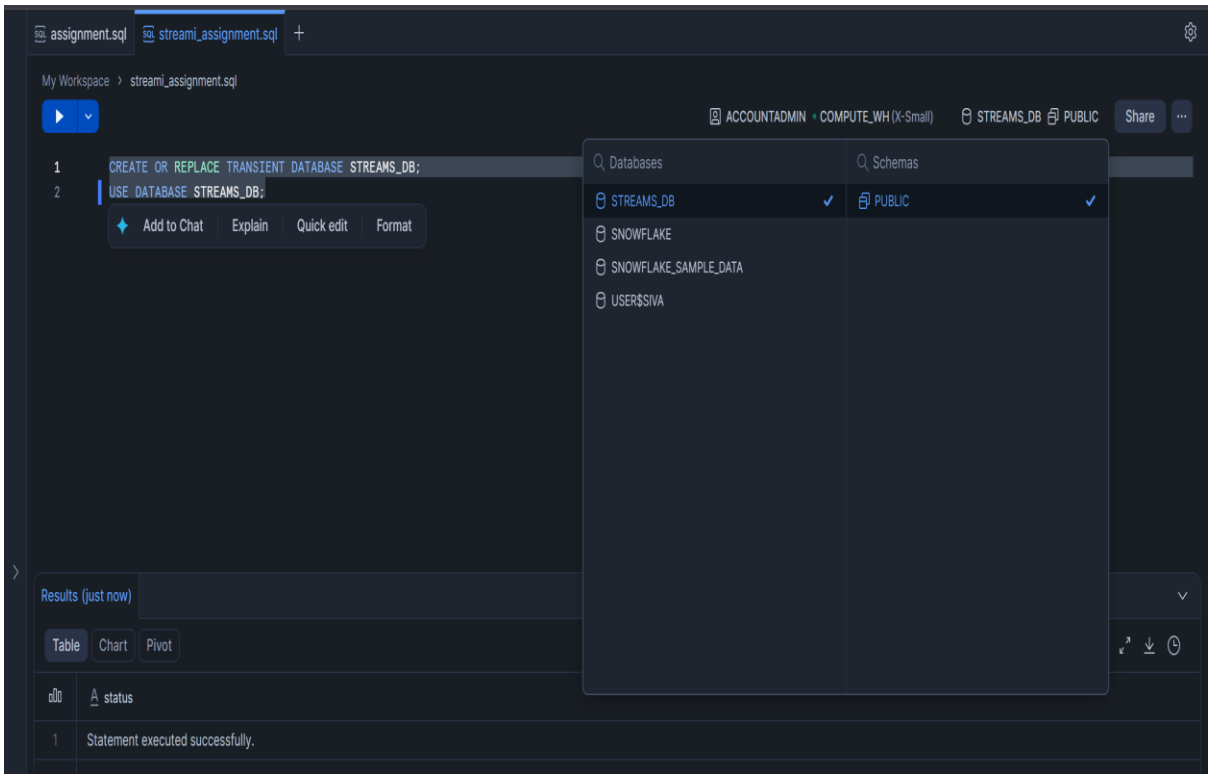
A retail company receives sales data in a staging table. The company wants to load this data into a final reporting table. After the initial load, only newly inserted or updated records should be processed using Snowflake Streams.

You need to create the database, tables, stream object, insert records, consume stream data, and update the final table using stream changes.

Task 1: Create Database

Create a transient database named:

STREAMS_DB



Task 2: Create Raw Sales Staging Table

Create a table named:

SALES_RAW_STAGING

The table should contain the following columns:

id

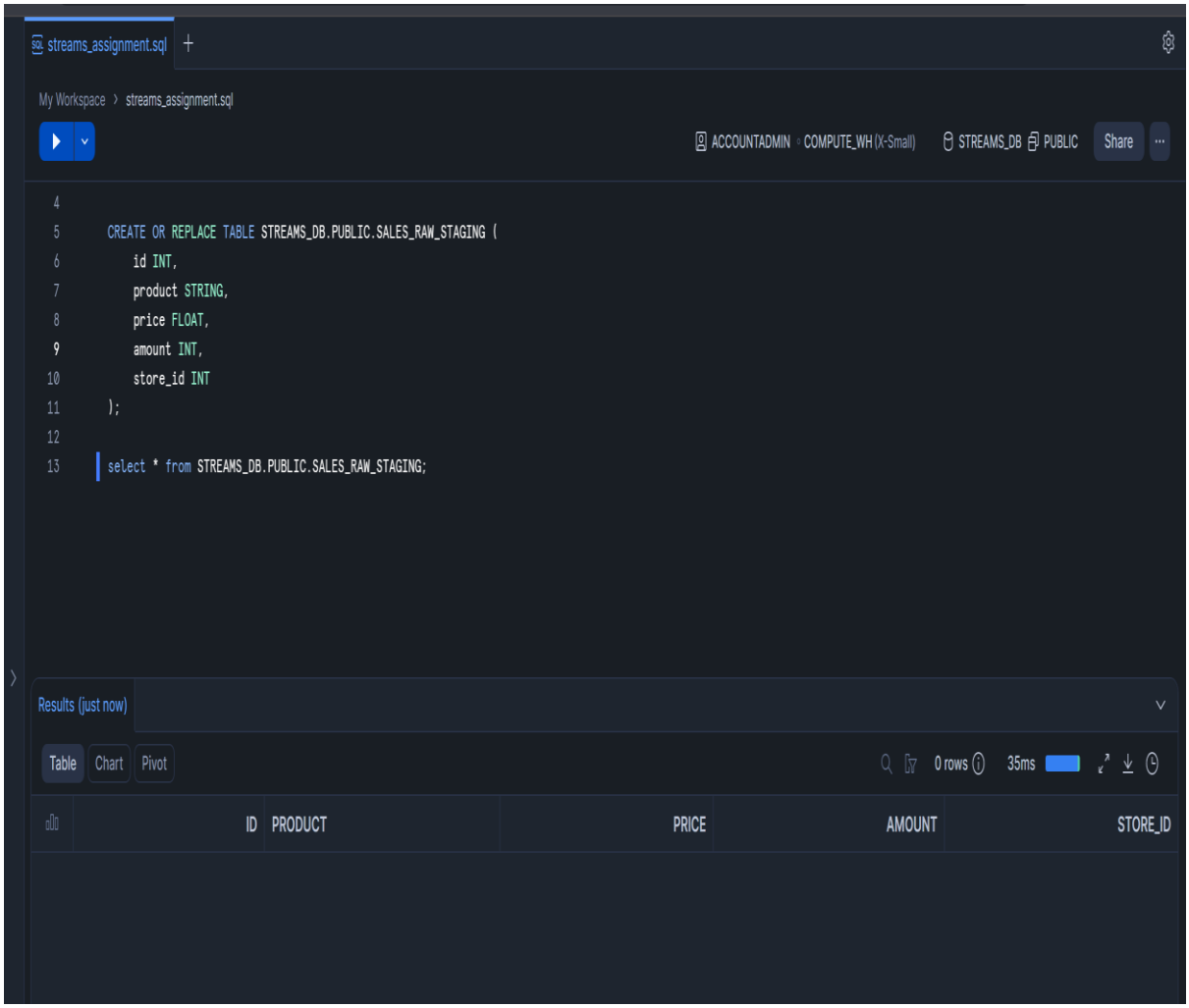
product

price

amount

store_id

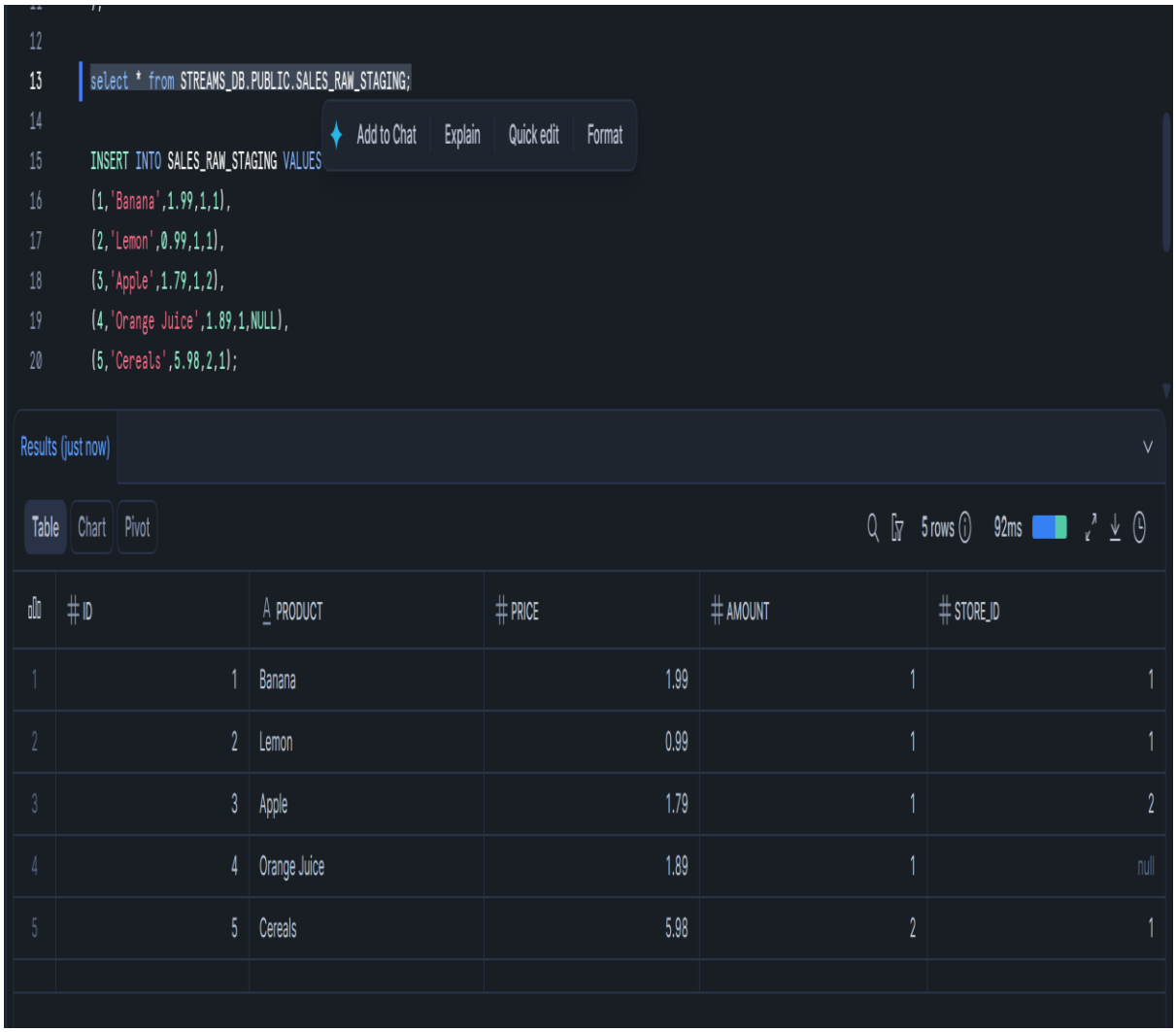
Use suitable data types.



Task 3: Insert Initial Sales Data

Insert the following five records into SALES_RAW_STAGING:

id	product	price	amount	store_id
1	Banana	1.99	1	1
2	Lemon	0.99	1	1
3	Apple	1.79	1	2
4	Orange Juice	1.89	1	1
5	Cereals	5.98	2	1



Task 4: Create Store Table

Create a table named:

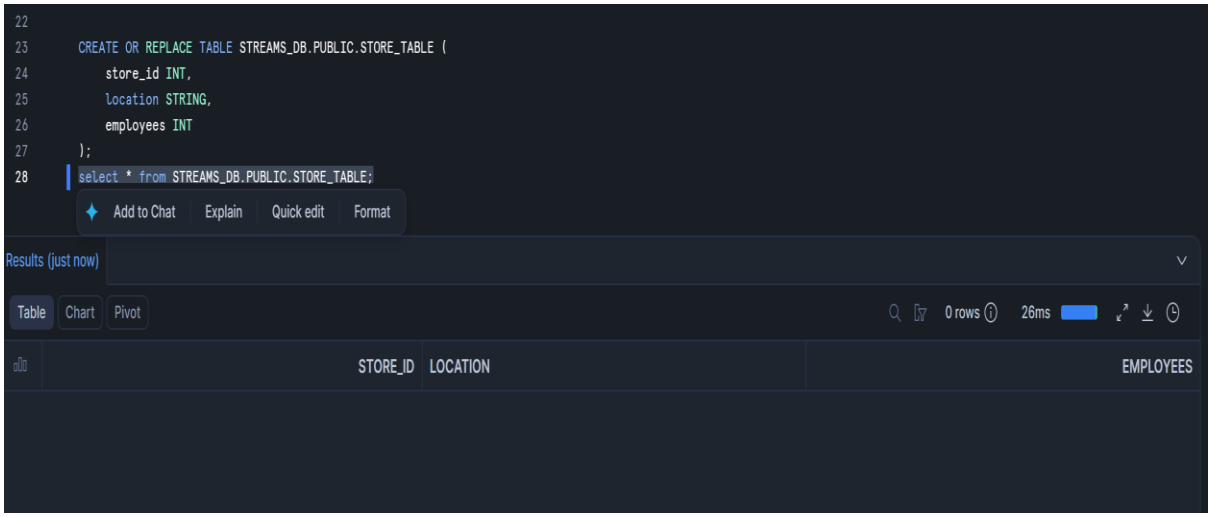
STORE_TABLE

The table should contain:

store_id

location

employees



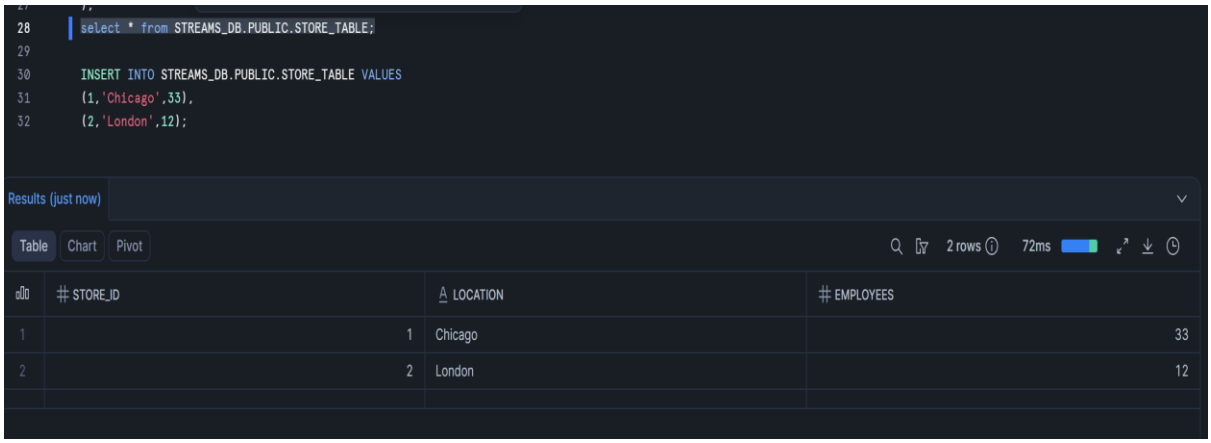
Task 5: Insert Store Data

Insert the following records into STORE_TABLE:

store_id locationemployees

1 Chicago 33

2 London 12



Task 6: Create Final Sales Table

Create a table named:

SALES_FINAL_TABLE

The table should contain:

id

product

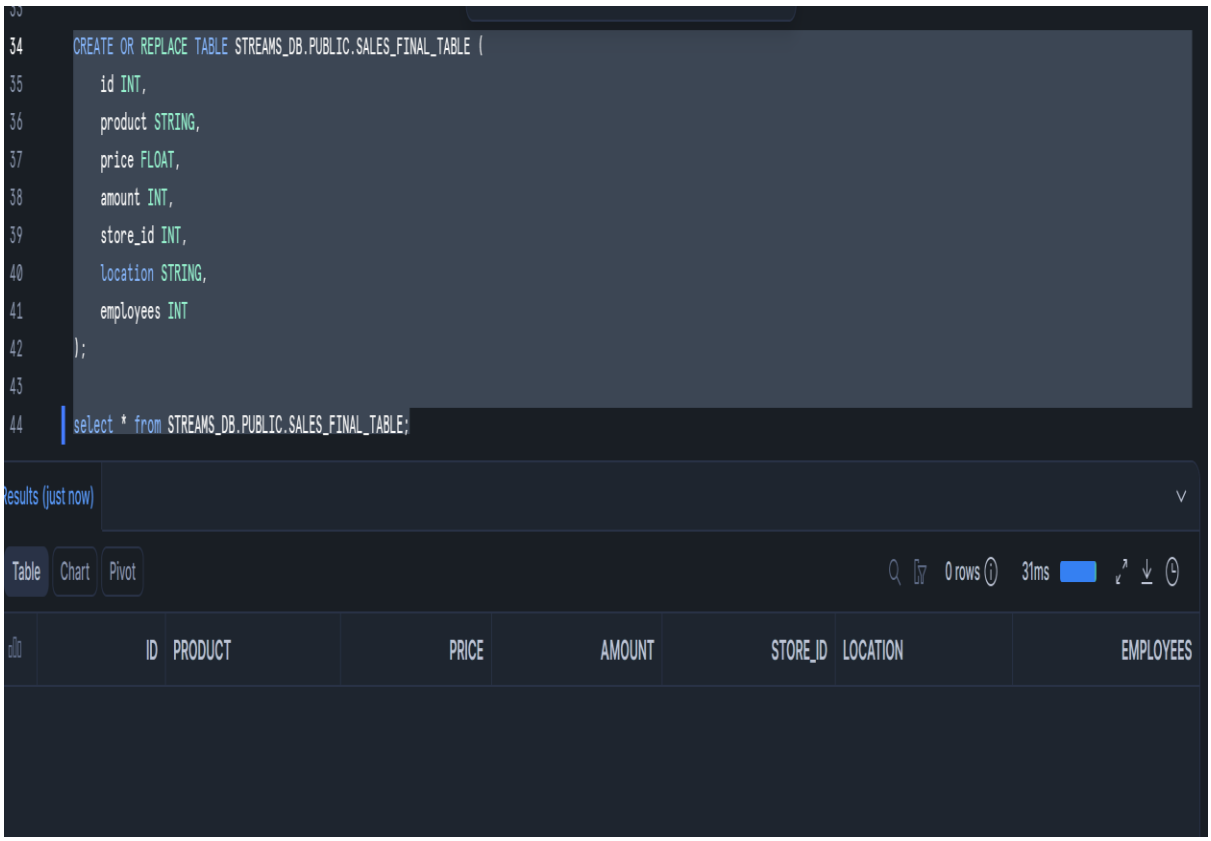
price

amount

store_id

location

employees



This table will store sales data along with store location and employee details.

Task 7: Perform Initial Full Load

Load data from SALES_RAW_STAGING into SALES_FINAL_TABLE.

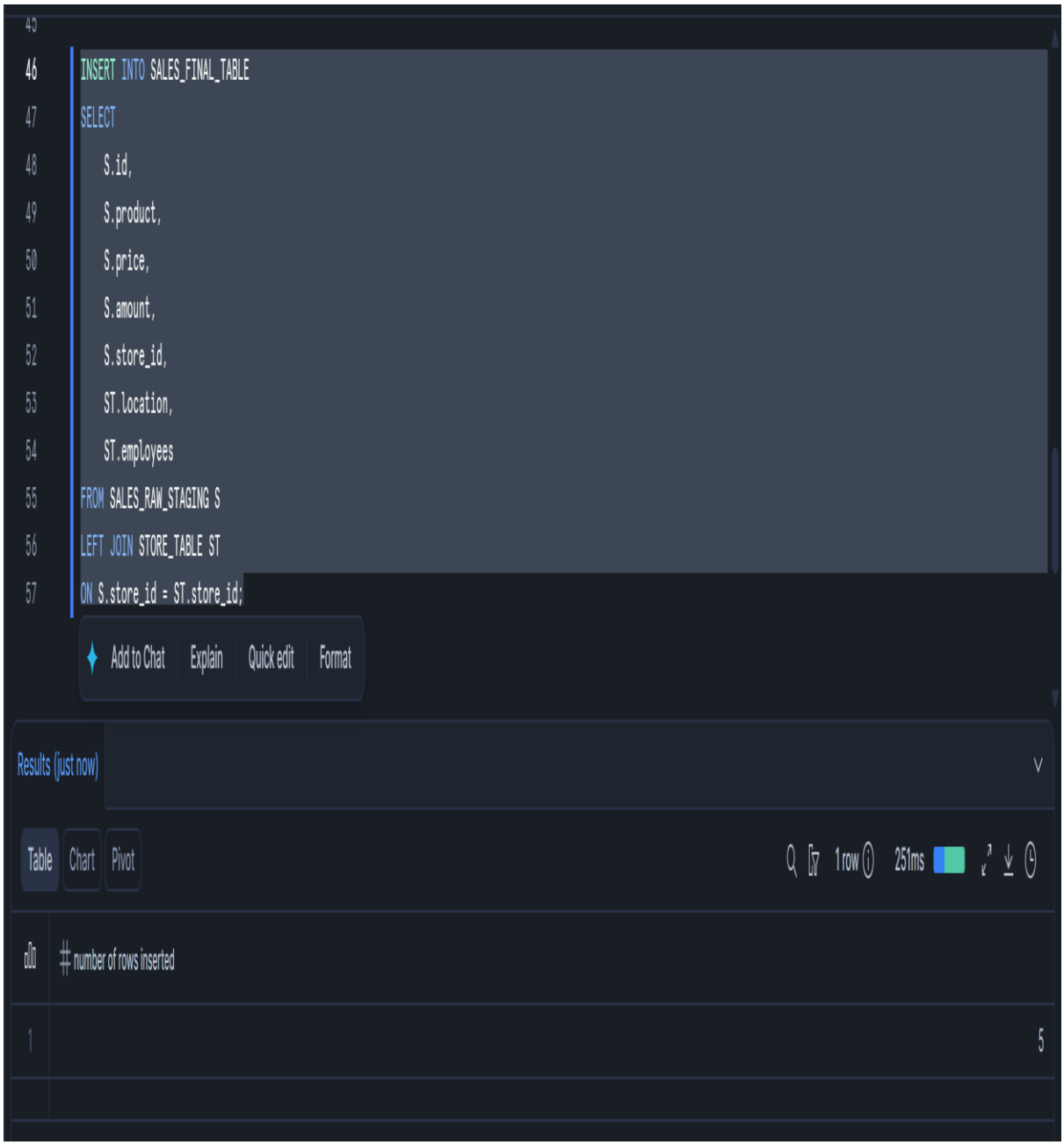
While loading, join:

SALES_RAW_STAGING

with:

STORE_TABLE

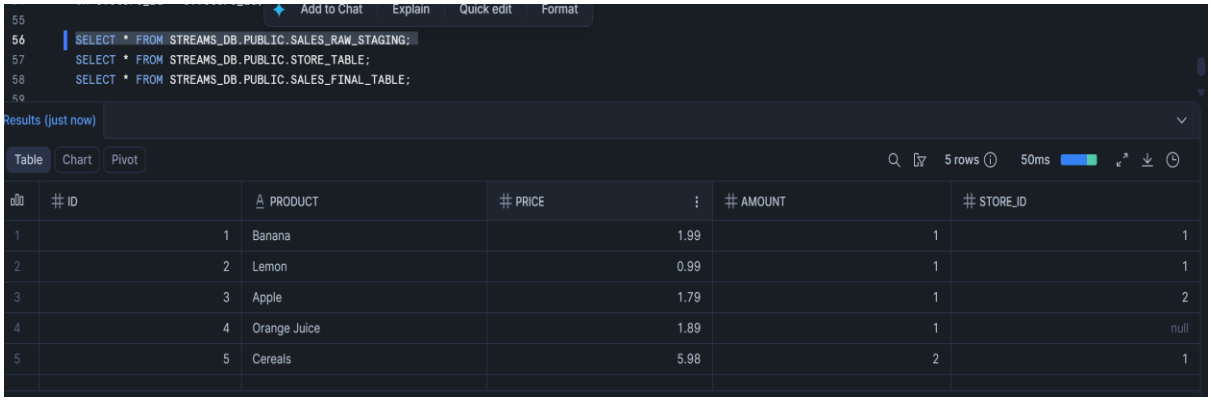
using STORE_ID.



Task 8: Verify Initial Load

Display data from:

SALES_RAW_STAGING;



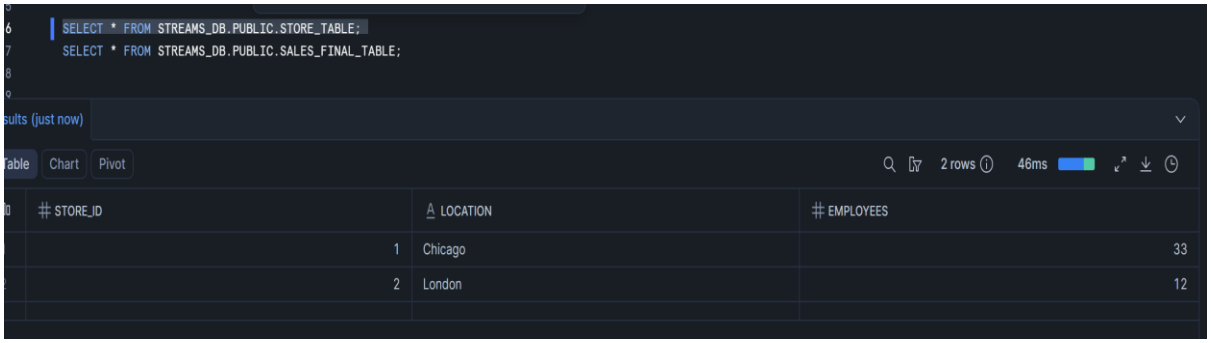
```
SELECT * FROM STREAMS_DB.PUBLIC.SALES_RAW_STAGING;
SELECT * FROM STREAMS_DB.PUBLIC.STORE_TABLE;
SELECT * FROM STREAMS_DB.PUBLIC.SALES_FINAL_TABLE;
```

Add to Chat | Explain | Quick edit | Format

Results (just now) — Table | Chart | Pivot — 5 rows — 50ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID
1	1	Banana	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1

STORE_TABLE;



```
SELECT * FROM STREAMS_DB.PUBLIC.STORE_TABLE;
SELECT * FROM STREAMS_DB.PUBLIC.SALES_FINAL_TABLE;
```

Table | Chart | Pivot — 2 rows — 46ms

	STORE_ID	LOCATION	EMPLOYEES
	1	Chicago	33
	2	London	12

SALES_FINAL_TABLE;

```
SELECT * FROM STREAMS_DB.PUBLIC.SALES_FINAL_TABLE;
```

Add to Chat | Explain | Quick edit | Format

Results (just now) — Table | Chart | Pivot — 5 rows — 25ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Banana	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33

Confirm that SALES_FINAL_TABLE contains 5 records.

Stream Implementation

Task 9: Create Stream Object

Create a stream named:

SALES_STREAM

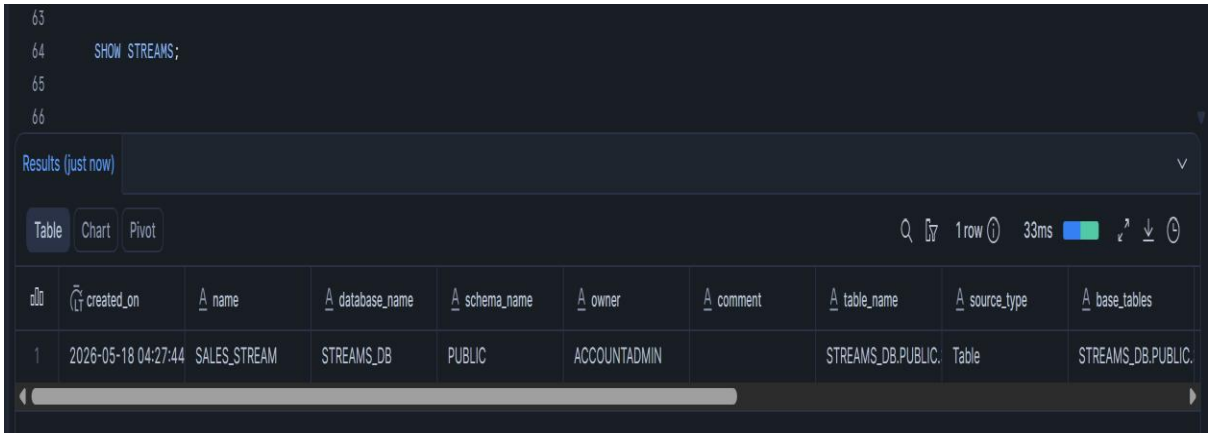
on top of:

SALES_RAW_STAGING

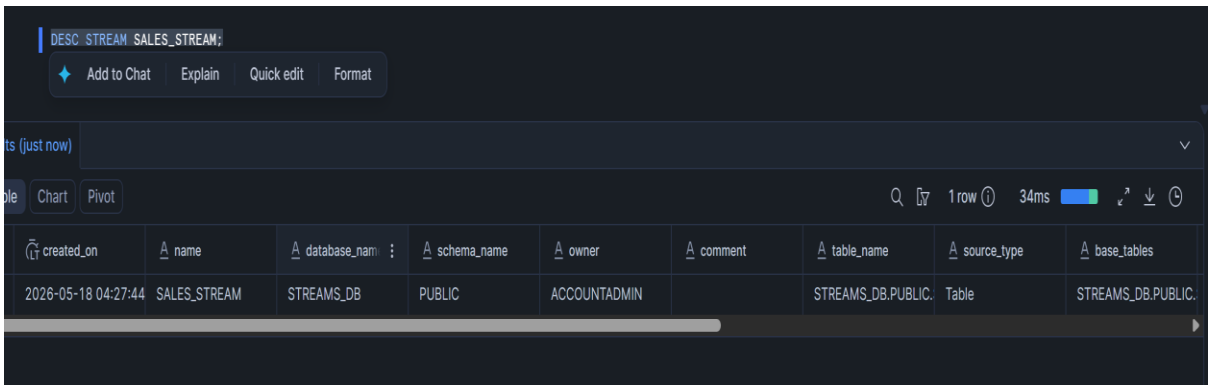
Task 10: View Stream Information

Execute commands to:

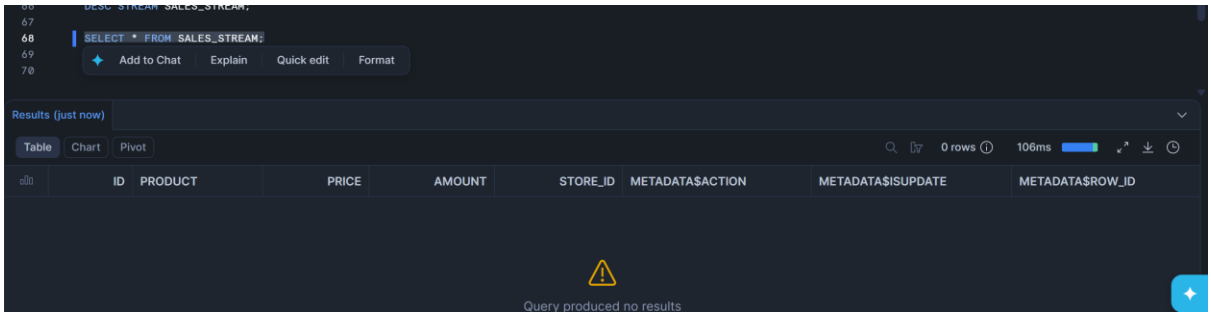
Show all streams.



Describe the SALES_STREAM.



Select data from SALES_STREAM.



Answer this question:

Why is SALES_STREAM empty immediately after creation?

BECAUSE STREAM CAPTURES ONLY NEW CHANGES AFTER CREATION

Insert Change Capture

Task 11: Insert New Sales Records

Insert the following two records into SALES_RAW_STAGING:

id	product	price	amount	store_id
6	Mango	1.99	1	2
7	Garlic	0.99	1	1

Task 12: Check Stream Data

```
INSERT INTO SALES_RAW_STAGING VALUES
(6,'Mango',1.99,1,2),
(7,'Garlic',0.99,1,1);

SELECT * FROM SALES_STREAM;
```

Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 2 rows — 119ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA$ACTION	METADATA$ISUPDATE	METADATA$ROW_ID
1	6	Mango	1.99	1	2	INSERT	FALSE	6b5670d0e2e3efe69dcbfbaa5c049adbd1666cf1
2	7	Garlic	0.99	1	1	INSERT	FALSE	d339500f0570498304767f5fa9d1530910e7ce06

Query SALES_STREAM.

Observe the new records and metadata columns:

METADATA$ACTION

METADATA$ISUPDATE

METADATA$ROW_ID

Answer this question:

What value do you see in METADATA$ACTION for newly inserted records?

INSERT

Task 13: Verify Source and Target Tables

Check data in:

SALES_RAW_STAGING;

```
SELECT * FROM SALES_RAW_STAGING;
```

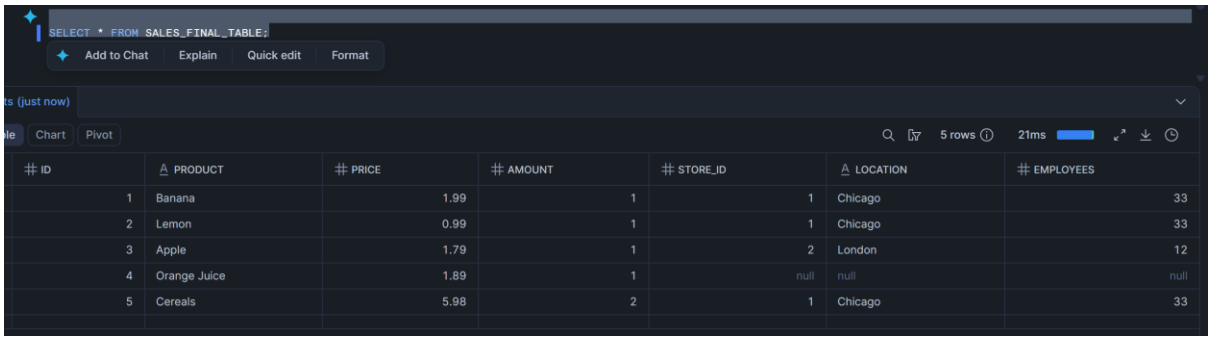
Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 7 rows — 62ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID
1	1	Banana	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1
6	6	Mango	1.99	1	2
7	7	Garlic	0.99	1	1

SALES_FINAL_TABLE;



ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	Banana	1.99	1	1	Chicago	33
2	Lemon	0.99	1	1	Chicago	33
3	Apple	1.79	1	2	London	12
4	Orange Juice	1.89	1	null	null	null
5	Cereals	5.98	2	1	Chicago	33

Answer this question:

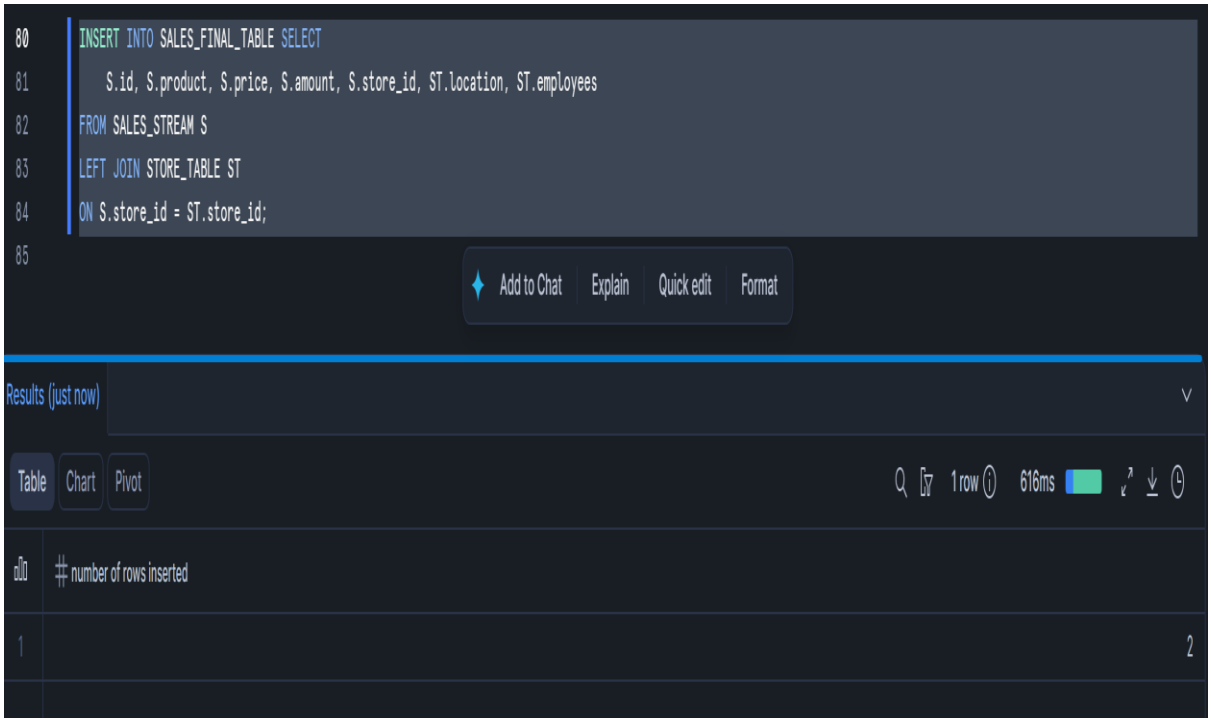
Are the new records available in SALES_FINAL_TABLE before consuming the stream?

NO NEW RECORDS ARE NOT FOUND IN FINAL TABLE

Task 14: Consume Stream Data

Insert new records from SALES_STREAM into SALES_FINAL_TABLE.

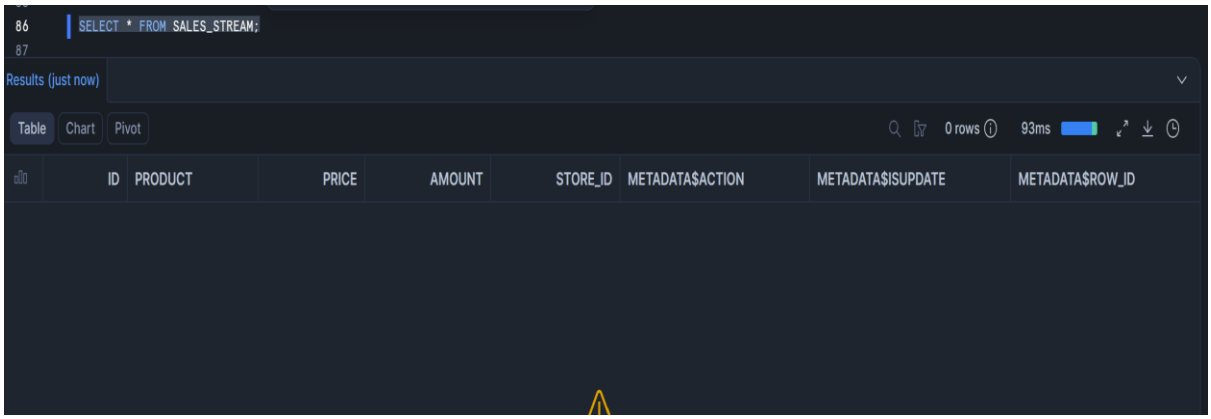
While inserting, join the stream with STORE_TABLE using STORE_ID.



Task 15: Verify Stream After Consumption

Query:

SELECT * FROM SALES_STREAM;



Answer this question:

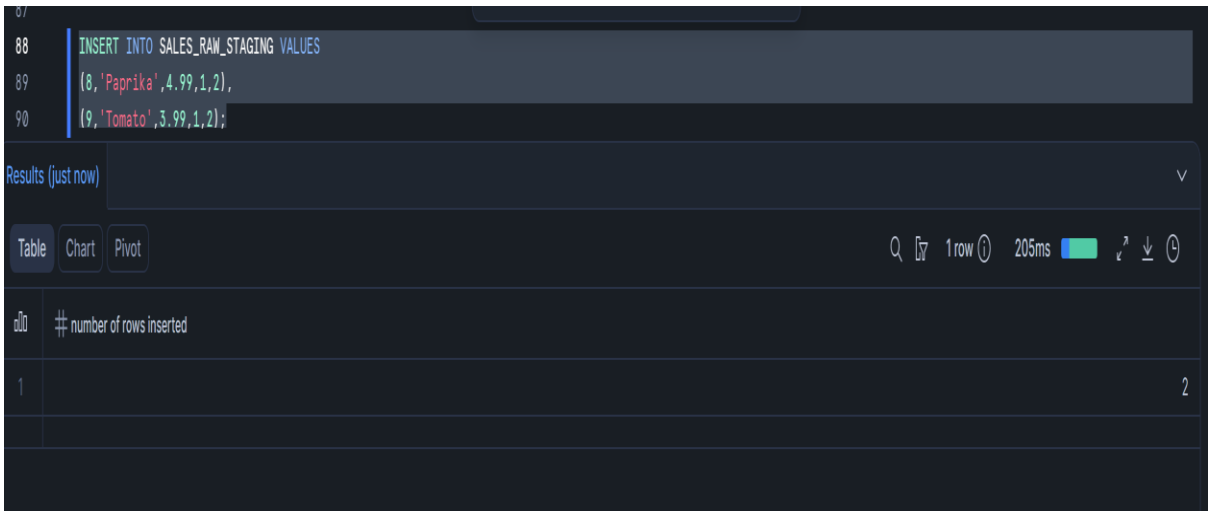
Why did the stream become empty after inserting data into SALES_FINAL_TABLE?

STREAM BECOMES EMPTY BECAUSE DATA IS CONSUMED

Task 16: Insert More Sales Records

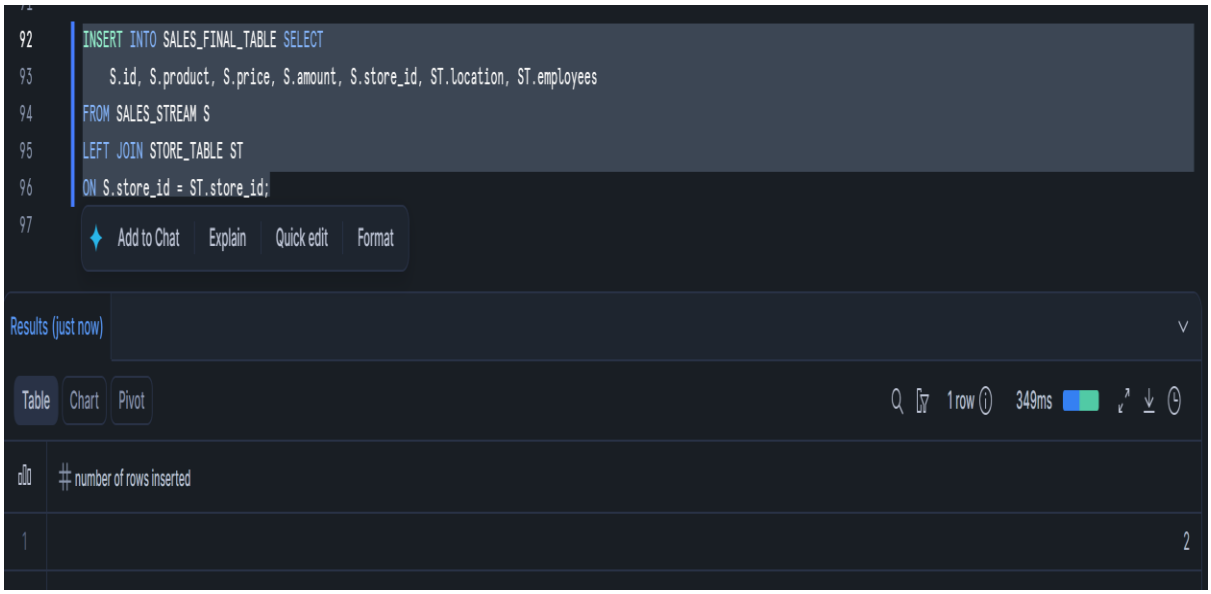
Insert the following two records into SALES_RAW_STAGING:

id	product	price	amount	store_id
8	Paprika	4.99	1	2
9	Tomato	3.99	1	2



Task 17: Consume Second Stream Data

Consume the newly captured stream records and insert them into SALES_FINAL_TABLE.



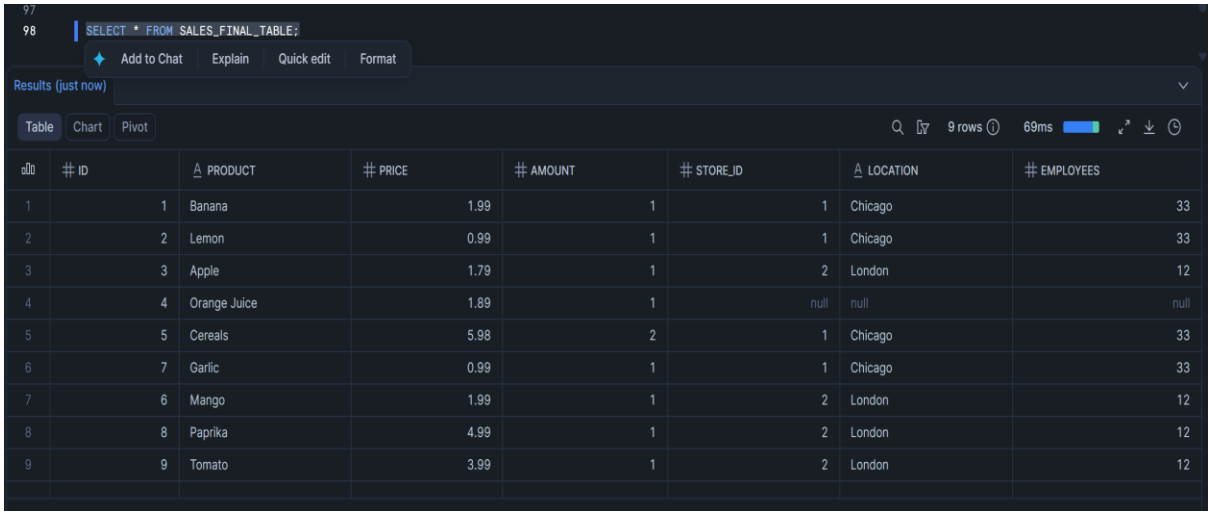
Task 18: Verify Final Insert Output

Display data from:

SALES_FINAL_TABLE;

SALES_RAW_STAGING;

SALES_STREAM;



	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Banana	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33
6	7	Garlic	0.99	1	1	Chicago	33
7	6	Mango	1.99	1	2	London	12
8	8	Paprika	4.99	1	2	London	12
9	9	Tomato	3.99	1	2	London	12

Answer:

How many records are now available in SALES_FINAL_TABLE? 9

Update Change Capture

Task 19: Check Existing Data Before Update

Display records from:

SALES_RAW_STAGING;

```
SELECT * FROM SALES_RAW_STAGING;
```

Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 9 rows — 57ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID
1	1	Banana	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1
6	6	Mango	1.99	1	2
7	7	Garlic	0.99	1	1
8	8	Paprika	4.99	1	2
9	9	Tomato	3.99	1	2

SALES_STREAM;

```
SELECT * FROM SALES_STREAM;
```

Add to Chat | Explain | Quick edit | Format

Results (just now)

Table | Chart | Pivot — 0 rows — 88ms

ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA$ACTION	METADATA$ISUPDATE	METADATA$ROW_ID

Task 20: Perform First Update

Update the product name:

Banana to: Potato

in SALES_RAW_STAGING.

```
UPDATE SALES_RAW_STAGING
SET product = 'Potato'
WHERE product = 'Banana';
```

Add to Chat | Explain | Quick edit | Format

Results (just now)

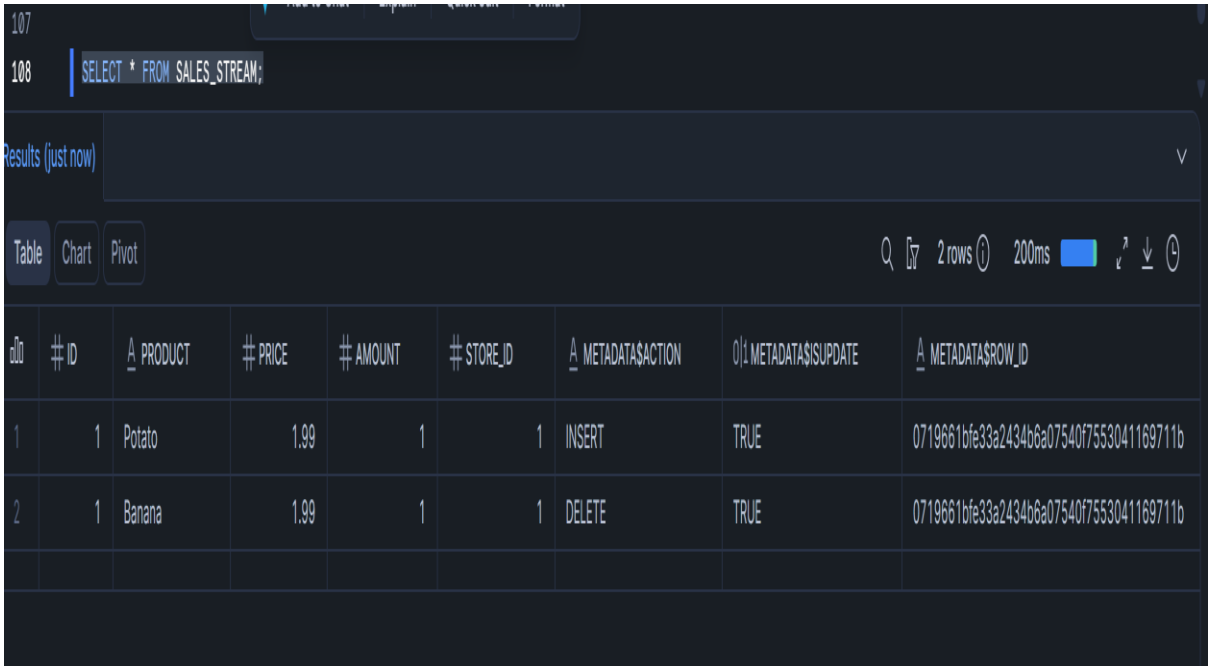
Table | Chart | Pivot — 1 row — 234ms

	number of rows updated	number of multi-joined rows updated
1	1	0

Task 21: Check Stream After Update

Query:

SELECT * FROM SALES_STREAM;



	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA$ACTION	METADATA$ISUPDATE	METADATA$ROW_ID
1	1	Potato	1.99	1	1	INSERT	TRUE	0719661bfe33a2434b6a07540f7553041169711b
2	1	Banana	1.99	1	1	DELETE	TRUE	0719661bfe33a2434b6a07540f7553041169711b

Answer:

How does Snowflake represent an UPDATE operation inside a stream?

INSERT

DELETE

Task 22: Consume Update Using MERGE

Use a MERGE statement to update the corresponding record in SALES_FINAL_TABLE.

Condition:

Match records using ID

Update only records where METADATA$ACTION = 'INSERT'

and METADATA$ISUPDATE = TRUE

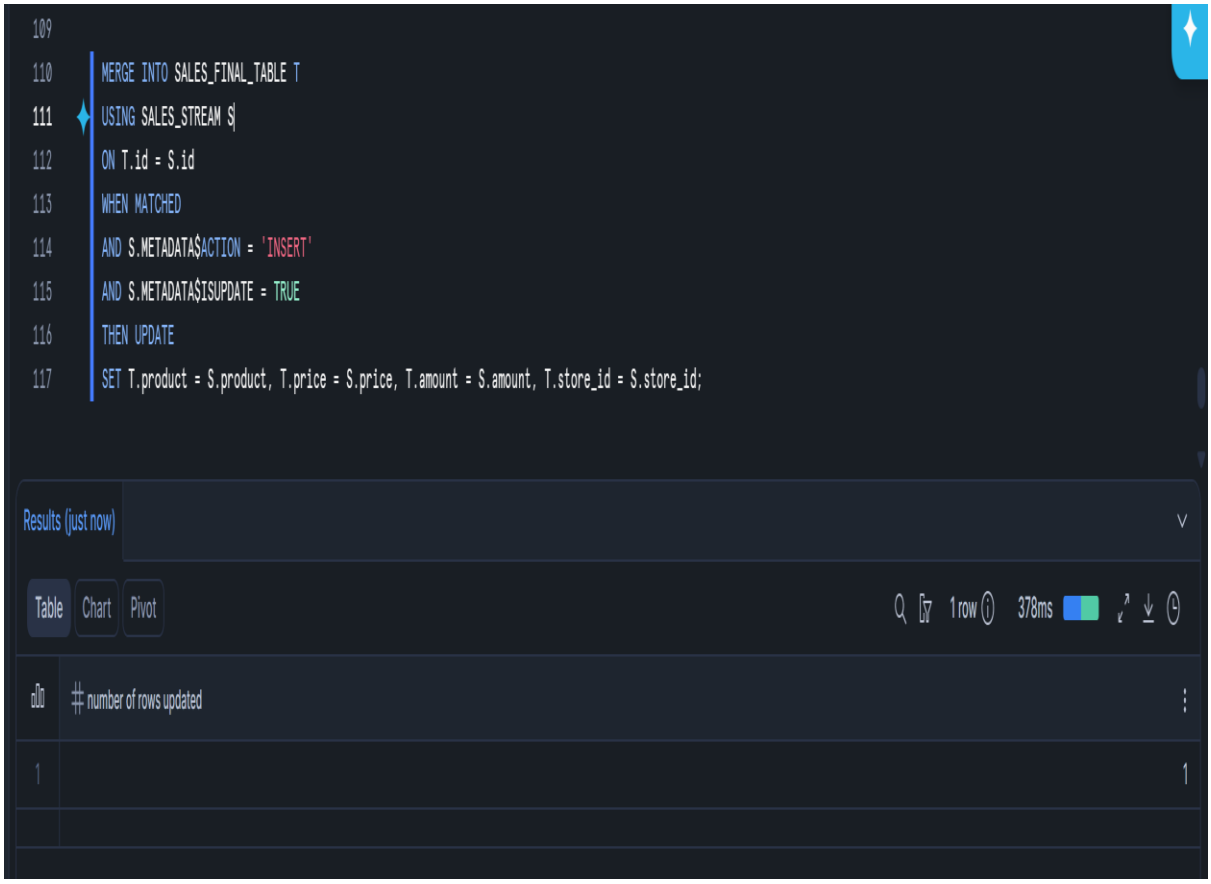
Update the following columns:

product

price

amount

store_id



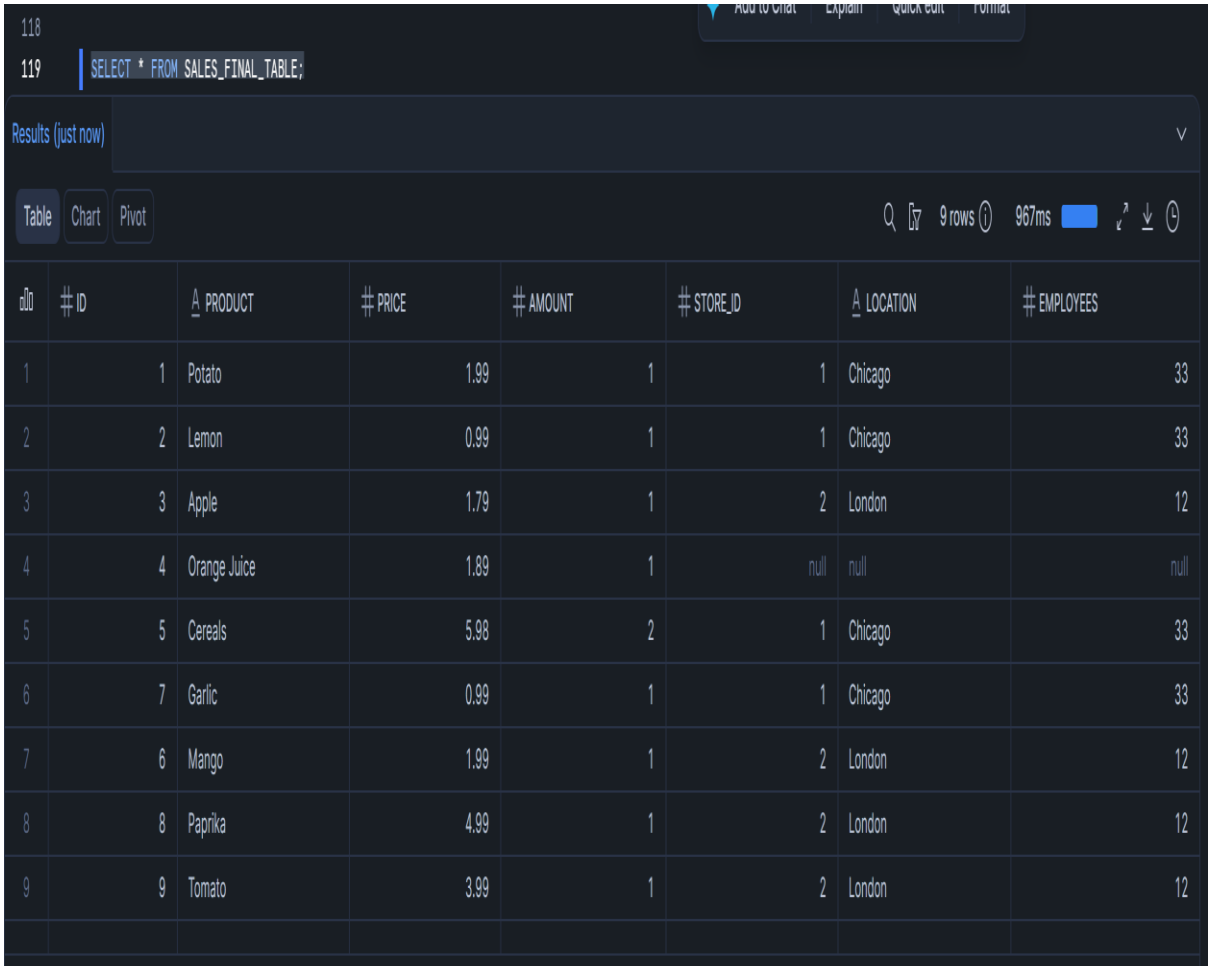
Task 23: Verify First Update

Display data from:

SALES_FINAL_TABLE;

SALES_RAW_STAGING;

SALES_STREAM;



```
118
119 SELECT * FROM SALES_FINAL_TABLE;
```

Results (just now)

Table | Chart | Pivot — 9 rows — 967ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Potato	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33
6	7	Garlic	0.99	1	1	Chicago	33
7	6	Mango	1.99	1	2	London	12
8	8	Paprika	4.99	1	2	London	12
9	9	Tomato	3.99	1	2	London	12

Answer:

Is Banana changed to Potato in SALES_FINAL_TABLE?

YSE-UPDATED

Second Update Change Capture

Task 24: Perform Second Update

Update the product name:

Apple

to:

Green apple

in SALES_RAW_STAGING.

```
123  UPDATE SALES_RAW_STAGING
124  SET product = 'Green apple'
125  WHERE product = 'Apple';
```

Results (just now)

Table | Chart | Pivot — 1 row — 524ms

	# number of rows updated	# number of multi-joined rows updated
1	1	0

Task 25: Consume Second Update Using MERGE

Use a MERGE statement to update the changed record in SALES_FINAL_TABLE.

```
109
110  MERGE INTO SALES_FINAL_TABLE T
111  USING SALES_STREAM S
112  ON T.id = S.id
113  WHEN MATCHED
114  AND S.METADATA$ACTION = 'INSERT'
115  AND S.METADATA$ISUPDATE = TRUE
116  THEN UPDATE
117  SET T.product = S.product, T.price = S.price, T.amount = S.amount, T.store_id = S.store_id;
118
```

Results (just now)

Table | Chart | Pivot — 1 row — 404ms

	# number of rows updated
1	1

Task 26: Verify Final Output

Display data from:

SALES_FINAL_TABLE;

```
129
130  Ctrl+I to generate
131  SELECT * FROM SALES_FINAL_TABLE;
```

Results (just now)

Table | Chart | Pivot — 9 rows — 23ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID	LOCATION	EMPLOYEES
1	1	Potato	1.99	1	1	Chicago	33
2	2	Lemon	0.99	1	1	Chicago	33
3	3	Green apple	1.79	1	2	London	12
4	4	Orange Juice	1.89	1	null	null	null
5	5	Cereals	5.98	2	1	Chicago	33
6	7	Garlic	0.99	1	1	Chicago	33
7	6	Mango	1.99	1	2	London	12
8	8	Paprika	4.99	1	2	London	12
9	9	Tomato	3.99	1	2	London	12

SALES_RAW_STAGING;

```
130
131  SELECT * FROM SALES_RAW_STAGING;
```

Results (just now)

Table | Chart | Pivot — 9 rows — 51ms

	ID	PRODUCT	PRICE	AMOUNT	STORE_ID
1	1	Potato	1.99	1	1
2	2	Lemon	0.99	1	1
3	3	Green apple	1.79	1	2
4	4	Orange Juice	1.89	1	null
5	5	Cereals	5.98	2	1
6	6	Mango	1.99	1	2
7	7	Garlic	0.99	1	1
8	8	Paprika	4.99	1	2
9	9	Tomato	3.99	1	2

SALES_STREAM;

```
133  SELECT * FROM SALES_STREAM;
```

Results (7 minutes ago)

Table | Chart | Pivot — 0 rows — 73ms

ID	PRODUCT	PRICE	AMOUNT	STORE_ID	METADATA$ACTION	METADATA$ISUPDATE	METADATA$ROW_ID

⚠

Answer:

Is Apple changed to Green apple in SALES_FINAL_TABLE?

YES-UPDATED